Duty at home

DM4 · One maximum and one equation · 60 min.

The function is considered f defined on $[0;+\infty[$ by $f(x) = (x+1)e^{-x}$. Its representative curve is noted $\mathcal{C}_f$.

1. Calculate $f(0)$ and limit of f in $+\infty$. Calculate $f'(x)$, then draw up the table of variations of f.

2. Calculate $f''(x)$. Study the convexity of f and determine the coordinates of the possible point Inflection of $\mathcal{C}_f$.

3. Demonstrate that the equation $f(x) = \frac{1}{2}$ admits a single solution α on $[0;+\infty[$.

4. Write a Python function that uses dichotomy on $[0;2]$ and returns two reals a and b such that $a \leq \alpha \leq b$ and $b - a \leq 10^{-3}$. Justify the choice of the initial interval.

5. Determine an equation of tangent to $\mathcal{C}_f$ at abscisse point 0. The position of the curve by in relation to this only tangent, is it sufficient to say that f is concave all over his field? Justify.